

ExoScout Preliminary Design Review



Team Name: ExoScout



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1 INTRODUCTION

1.1 Team organisation and roles

Team consists of 6 students from V High School in Bielsko-Biała, where we pursue extended programmes in math, physics and computer science. Although all of us realise the same extended subjects we present a wide variety of interdisciplinary interests and abilities. We see the CanSat competition as a great opportunity to broaden our knowledge and get experience in this type of team work. On top of that we are supported by our teacher and one of our older schoolmates, who is our mentor.

[REDACTED] - teacher:

- Background: [REDACTED]
- Field of work: Supervision of the team's progress.

[REDACTED] - mentor:

- Background: [REDACTED]
- Field of work: Consultation.
- Hours dedicated weekly: 1 hour for consultations

Karol Synowiec - team leader:

- Background: Interested in math, physics and engineering. He likes understanding how machines work and is eager to explore new science achievements especially those concerning space exploration and physics related to it. He participates in contests such as Math Olympiad, Physics Olympiad, Diament AGH and more.
- Fields of work: CanSat 3D modelling in Fusion 360, electronics and team leader
- Hours dedicated weekly: 10 hours

[REDACTED] - team member:

- Background: [REDACTED]

¹ <https://spacecamp.no/>

- [REDACTED]
- [REDACTED]
- Field of work: Software concerning sensors
 - Hours dedicated weekly: 12 hours

[REDACTED] - team member:

- Background: [REDACTED]
[REDACTED]
[REDACTED]
[REDACTED]
[REDACTED]
- Field of work: Mechanical system development and design
- Hours dedicated weekly: 5 hours

[REDACTED] - team member:

- Background: [REDACTED]
[REDACTED]
[REDACTED]
[REDACTED]
- Fields of work: Recovery system, CanSat assemblment.
- Hours dedicated weekly: 10 hours

[REDACTED] - team member:

- Background: [REDACTED]
[REDACTED]
[REDACTED]
[REDACTED]
[REDACTED]
- Fields of work: software concerning soil intake, electronics.
- Hours dedicated weekly: 5 hours

[REDACTED] - team member:

- Background: [REDACTED]
[REDACTED]
[REDACTED]



- Fields of work: outreach, promotion
- Hours dedicated weekly: 5 hours

1.2 Mission objectives

1.3 Our vision

In our project we want to create CanSat which will act as a "scout" for a mission looking for a planet that can be settled by humans. After first assessments of a planet's potential habitability from space, a CanSat like device will be dropped on the planet's surface and will carry out further measurements of factors crucial to the life of the humankind. Data collected from primary and secondary missions will be used with a view to analyse multiple factors, which are considered to be vital for planetary habitability.

1.4 Primary mission

Primary mission is to measure air temperature and air pressure and to transmit it as a telemetry once per second to the ground station. Data collected from it will be used to assess if water can be liquid on the planet's surface². Temperature limit for life³ is in the range of 0 to 113 degrees Celsius, which will be also evaluated based on those measurements.

1.5 Secondary mission

Secondary mission will collect more specific data about the planet. In order to do that we divide it into the two main stages:

1. During fall CanSat will measure factors concerning planetary habitability:
 - Gravitational acceleration - Too high value of it makes any motion harder, blood can't freely flow and bones can be broken due to overload⁴⁵. On the other hand low gravity implies low escape velocity, it results in molecules leaving the planet more easily and destroying its atmosphere⁶.
 - Cosmic radiation (UV) - Radiation is both helpful and harmful⁷. UV radiation takes part in biosphere processes but also affects human health. Too much of it can result in cancer but it is crucial for vitamin D3 synthesis in the body.
 - Magnetic field - without proper magnetic field the planet is not protected from stellar wind and cosmic radiation, which destroys its atmosphere and puts life at risk of being bombarded with ionized particles⁸.

² Liquid water importance from abstract of: <https://pubmed.ncbi.nlm.nih.gov/11539468/>

³ Table 1 on page 185 http://www.geology.wisc.edu/astrobiology/docs/Lammer_et_al_2009_Astron_Astro_Rev.pdf

⁴ <https://www.astronomy.com/science/whats-the-maximum-gravity-we-could-survive/>

⁵ <https://arxiv.org/pdf/1808.07417.pdf>,

⁶ extract about mass and size and hence planets gravity from https://en.m.wikipedia.org/wiki/Planetary_habitability

⁷ <https://www.cdc.gov/nceh/features/uv-radiation-safety/index.html>

⁸ Significance part from: https://en.wikipedia.org/wiki/Earth%27s_magnetic_field



- Atmosphere O₂ content - oxygen is vital to life as we know it. The reduction of oxygen provides one of the largest free energy release per electron transfer⁹, so it is a great source of energy for complex life.
- 2. Vertical landing and taking a soil sample with Archimedes screw-like device¹⁰ to measure its humidity:
 - Analysis of soil helps in the search for past water activities. Mineralogy of it is important in determining the processes that created the planet. Soil is widely analysed by rovers (f.e. on Mars)¹¹.

1.6 Assessment

In conditions of the launch campaign on Earth we will test the measurements correctness by comparing them with those measured by more sophisticated devices. From the mechanical side we want to test if we can make CanSat land vertically by reducing its speed and using side stabilizers.

⁹ <https://earthscience.rice.edu/wp-content/uploads/2015/08/Catling-Astrobiology-2005.pdf>

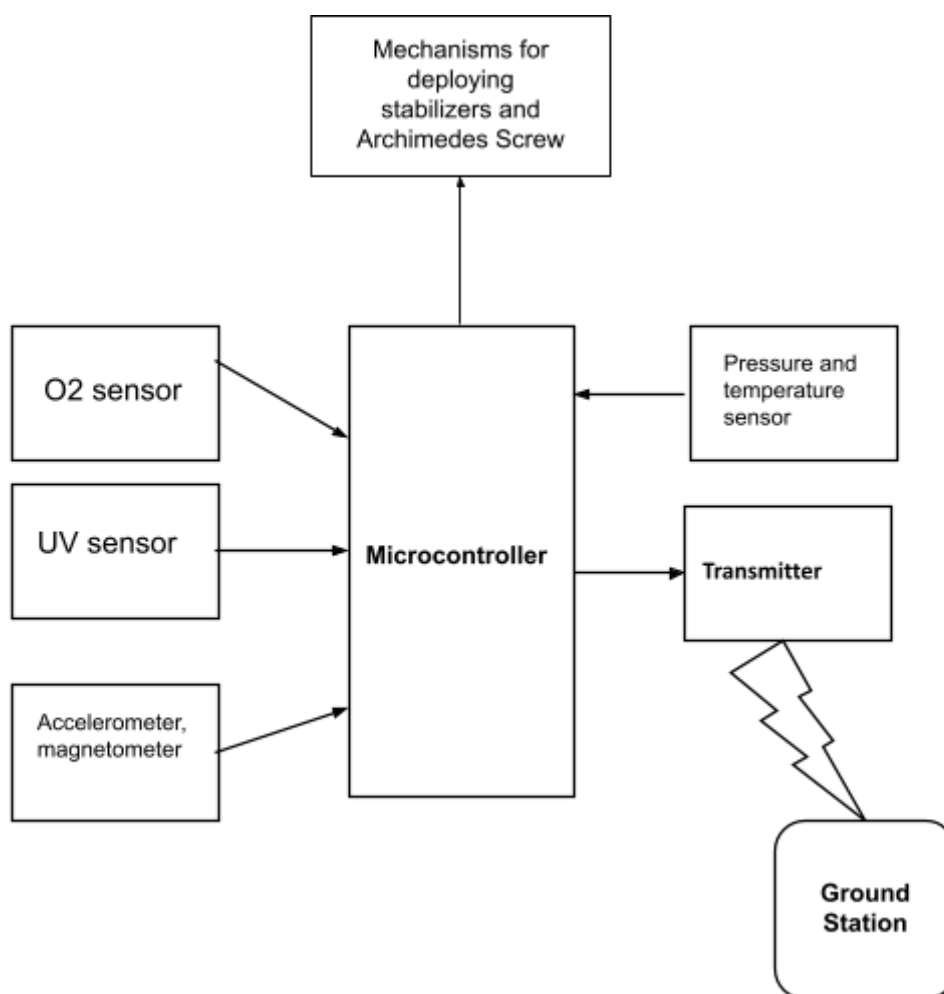
¹⁰ https://en.wikipedia.org/wiki/Archimedes%27_screw

¹¹ <https://mars.nasa.gov/resources/4910/inspecting-soils-across-mars/>

2 CANSAT DESCRIPTION

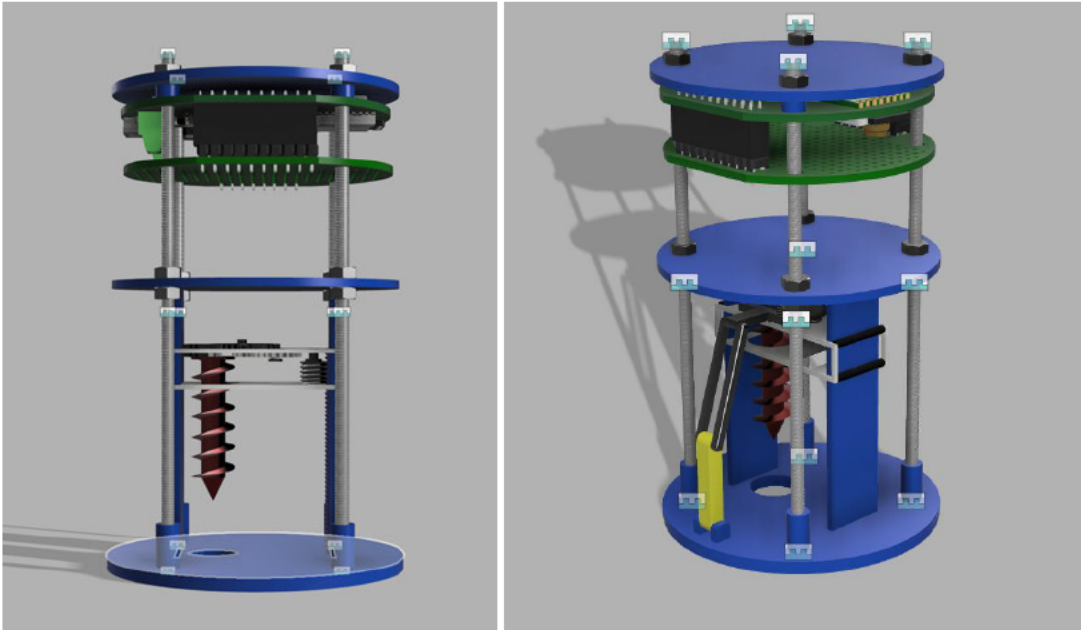
2.1 Mission overview

After launch from a rocket at a destined altitude (1500m-2500m) CanSat will deploy a parachute and side stabilizers. During the fall the device will start its measurements and reach terminal velocity of around 6 m/s in order to land vertically which will enable us to collect soil with Archimedes Screw-like mechanism after landing.

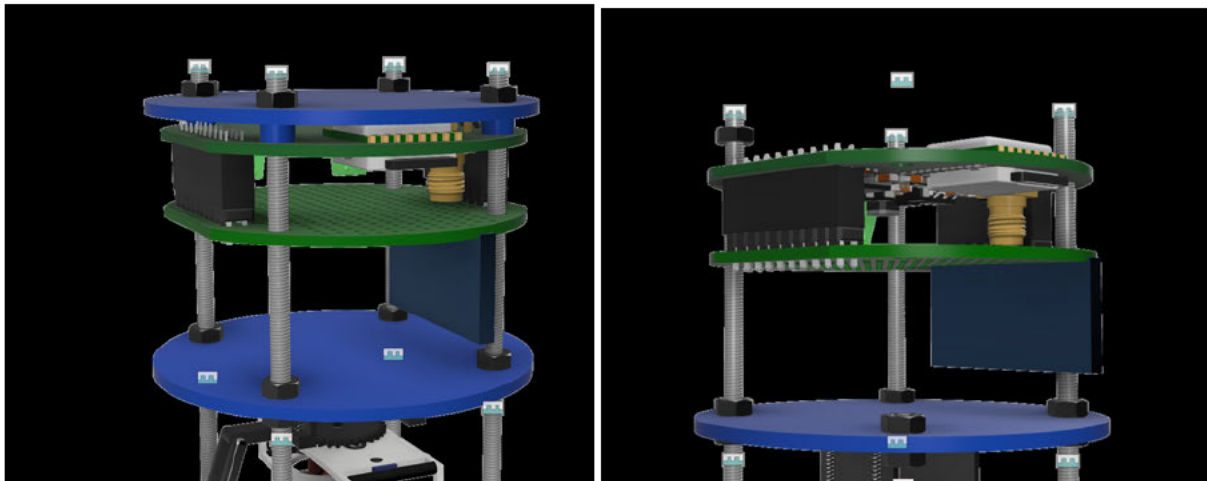


2.2 Mechanical/structural design

CanSat's cover and some internal parts will be printed on a 3D printer with ABS filament. It will be rigidly mounted on stainless rods with nuts. CanSat will be divided into two parts as shown in the render without cover:

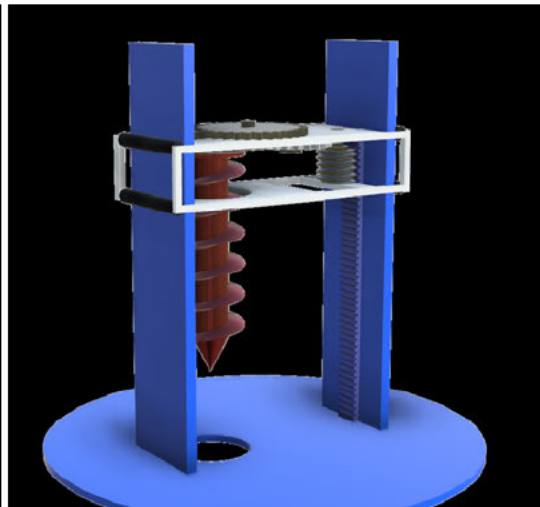
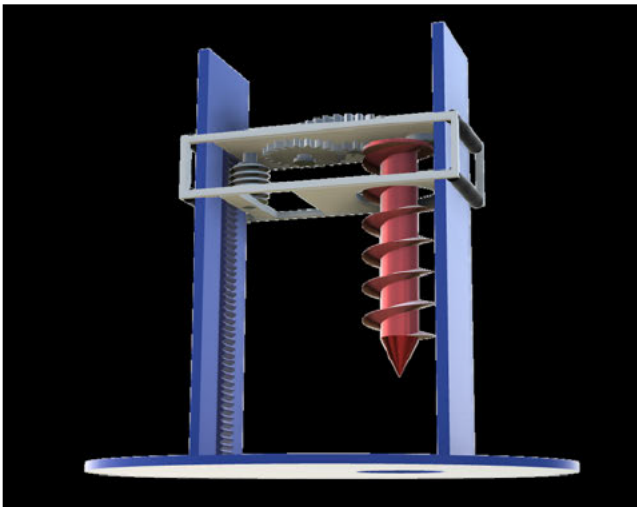
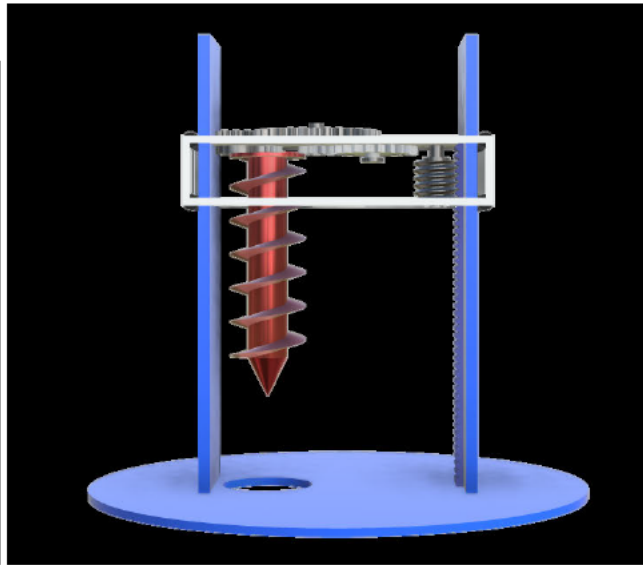
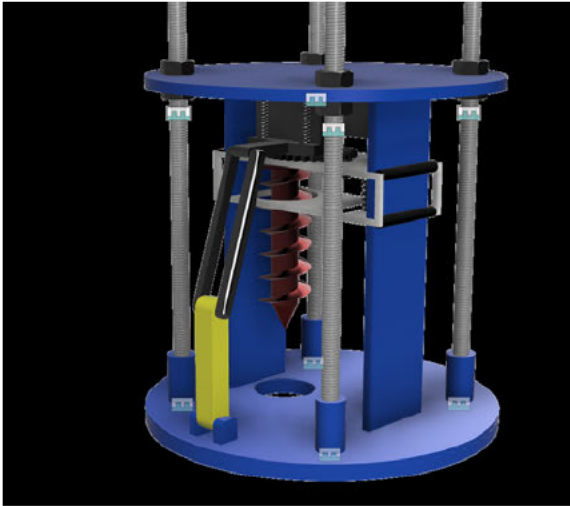


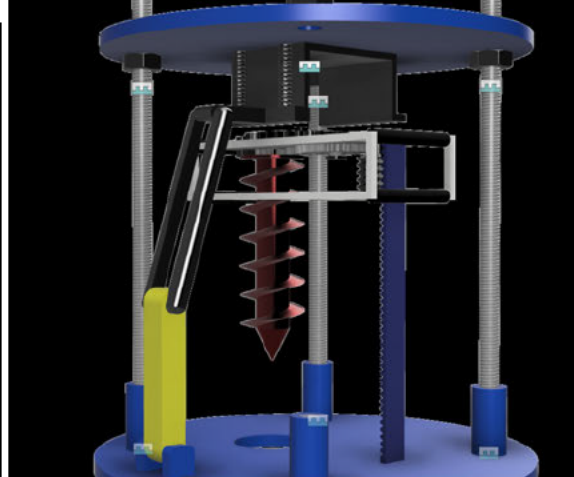
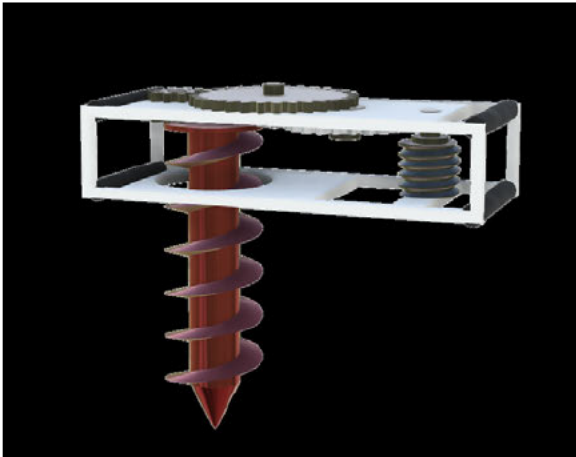
Upper part will be a place for a computer, battery, prototype board and sensors



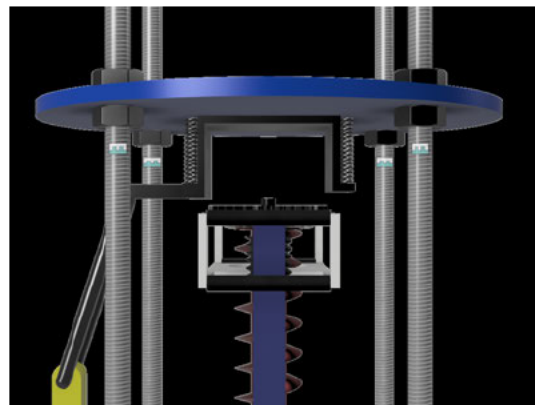
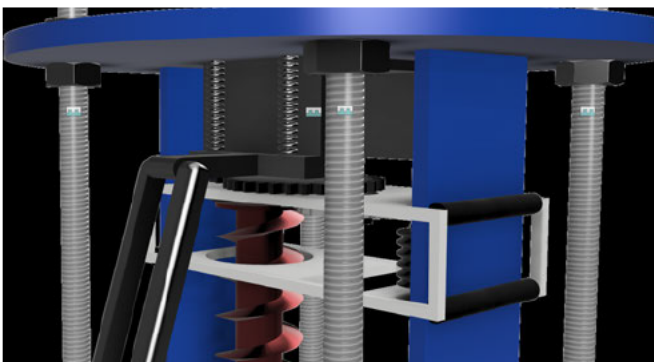


Lower part will be a place for a mechanisms to collect soil and deploy stabilizers, which looks like that:

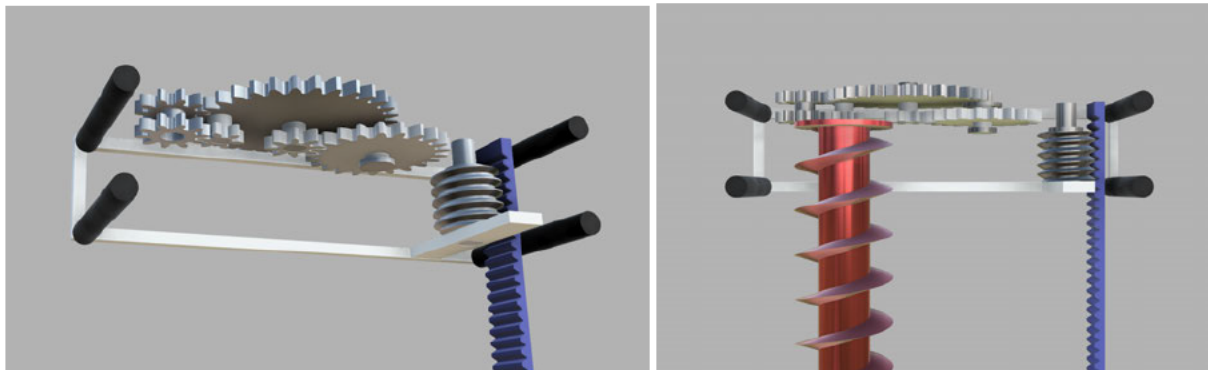




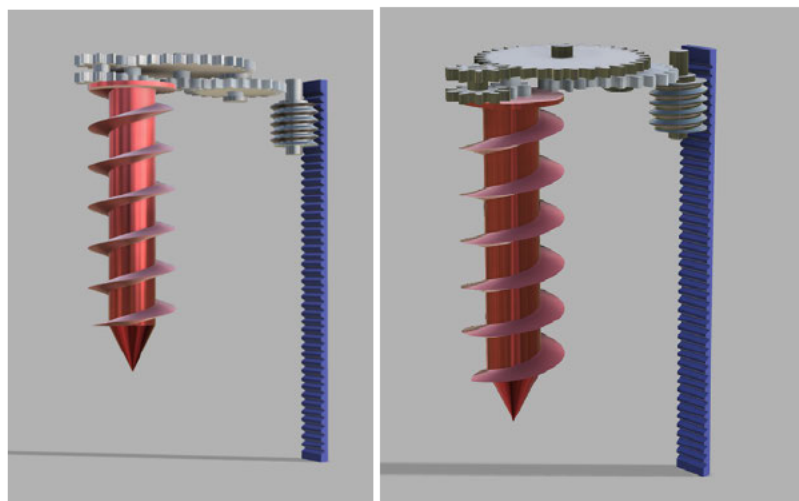
Whole drilling device with the motor and soil humidity control unit is suspended on a moving elevator. Thanks to a clever gearing mechanism both drill rotation and trolley lowering are done simultaneously, at suitable RPM. What is more, the mechanism for deploying stabilizers is also powered without use of any additional motor, due to adaptation of the spring system. Being still in the air, the drilling module will descend by 15 mm, thus enabling the set of springs to stretch downwards together with a specially shaped element that does not interfere and restrict the motor space (note that on the renders this is an empty space). 4 stabilizers will be mounted to the yellow space of each arm, connected via double rod to the platform housing previously mentioned springs. With the help of gravity, they will push the leg down, effectively unfolding the landing gear. It is important to mention that after deployment, landing gear consisting of stabilizers will be suspended thanks to a use of springs, which will be able to move back and forth freely, softening the landing and offering additional protection.



High torque electric motor will be installed in a space 16x32x14 mm, above 32-teeth gear, transferring power on it. Then the power will be distributed among 2 systems: drill rotation and elevation mechanism. The first one is geared up, by a set of gears to provide high RPM, necessary for efficient drill operation. The latter one on the other hand is geared down, to make a slow, but steady descent, with enough power to push the drill against solid ground. The input power is transferred by a transmission system made up of a set of gears and rubber band, directly propelling the worm gear.



Use of such an unusual solution is intentional. When the cart reaches the lowest position, further descent would normally put huge stress on the CanSat body, potentially causing cracks in the base plate. However with a use of properly adjusted rubber band, it will start to slip after encountering assumed resistance, thus enabling further drill operation, with a cut in power supply to the worm gear. This ingenious solution does not create a need for an additional motor to power the lift mechanism independently, effectively lowering both overall mass and cost of the CanSat.





Mass estimated by Fusion 360, after applying the right materials, is around 204g. After adding sensors, side stabilisers, parachute and soil intake mechanism mass will be in the range of 300-350g.

2.3 Electrical design

Electrical design includes:

- Computer - from CanSat kit
- Proto board - from CanSat kit
- Battery - capacity will be estimated later
- Primary mission sensors - temperature and pressure sensors from CanSat kit
- Secondary mission sensors:
 - UV sensor e.g:
<https://botland.store/sensors-of-light-and-color/19004-ltr390-uv-ultraviolet-light-sensor-stemma-qt-5903351249423.html>
 - O2 sensor e.g:
<https://botland.store/gravity-gas-dust-sensors/20981-gravity-o2-sensor-i2c-uart-factory-calibrated-dfrobot-sen0465-5904422383299.html>
 - Accelerometer and magnetometer:
<https://botland.store/9dof-imu-sensors/17709-icm-20948-9dof-3-axis-accelerometer-gyroscope-and-magnetometer-i2csbi-qwiic-adafruit-4554-5904422378677.html>
 - Servomotor:
<https://botland.com.pl/silniki-micro-pololu-seria-hp-high-power/246-silnik-hp-1001-pololu-1101-5904422306106.html>

2.4 Software design

In primary mission software should plot measured temperature and pressure as a point on phase diagram¹² in order to check if water is liquid in these conditions. In the secondary mission software will check if measured factors are in range optimal for life and send information to ground stations. Data should be live displayed on laptops with use of software like Grafana¹³. Program will also launch side stabilisers after drop from the racket and start the Archimedes screw after landing.

2.5 Recovery system¹⁴

Parachute will be made out of nylon and will have a hemispherical shape. It will be attached to the CanSat with four lines. Its area can be obtained from the equation for equilibrium of forces:

¹² <https://openstax.org/books/chemistry-2e/pages/10-4-phase-diagrams>

¹³ <https://grafana.com/>

¹⁴ <https://cansat.esa.int/design-your-parachute/>



$$mg = \frac{1}{2}C_D\rho Av^2$$

Where: m - mass of the CanSat, g - Earth acceleration, C_D - drag coefficient of parachute, ρ - density of air and v is terminal velocity.

After some careful conversions we get:

$$A = \frac{2mg}{C_D\rho v^2}$$

Substituting values:

$g=9.81 \text{ m/s}^2$, $C_D=0.62$, $\rho=1.204 \text{ kg/m}^3$, $v=6 \text{ m/s}$, m in range of 0.3 to 0.35 kg

We get:

$$0.219 \text{ m}^2 \leq A \leq 0.255 \text{ m}^2$$

Shape of hemispherical parachute:

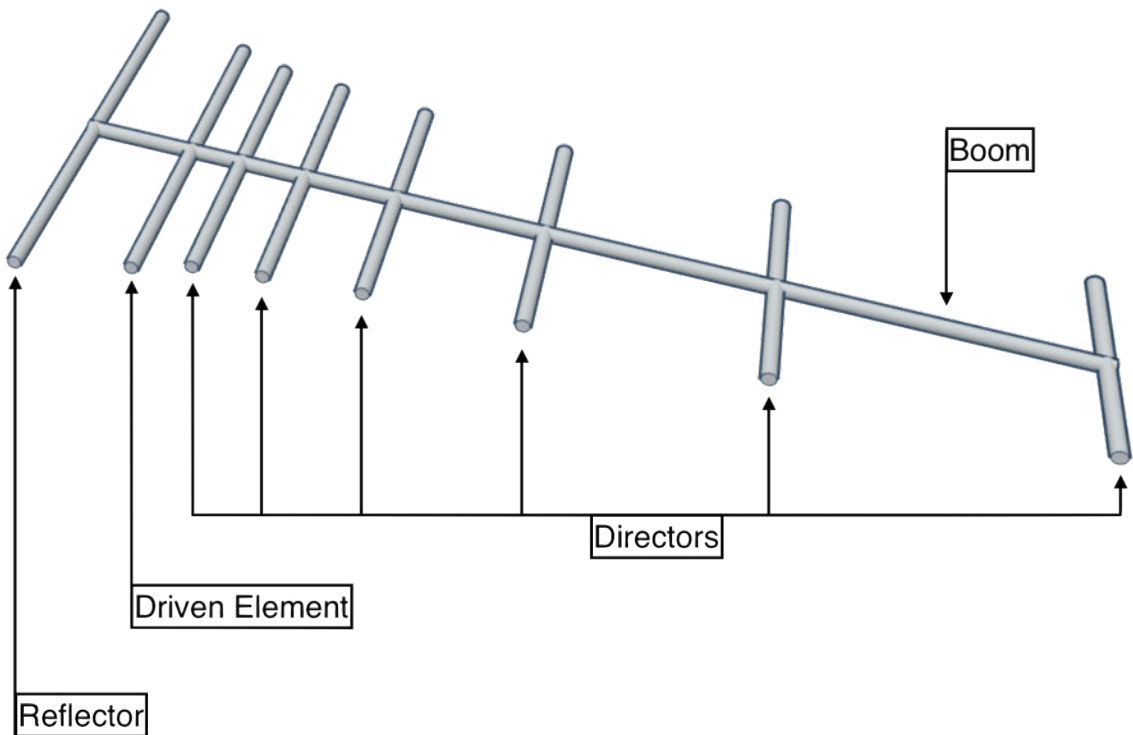




2.6 Ground support equipment¹⁵

Data from the CanSat will be transmitted to the ground station. Equipment concerned with these will consist of:

- Yagi-Uda antenna:



- Computer to display received data
- Adafruit Feather M0 RFM96 LoRa Radio for communication of antenna and the computer

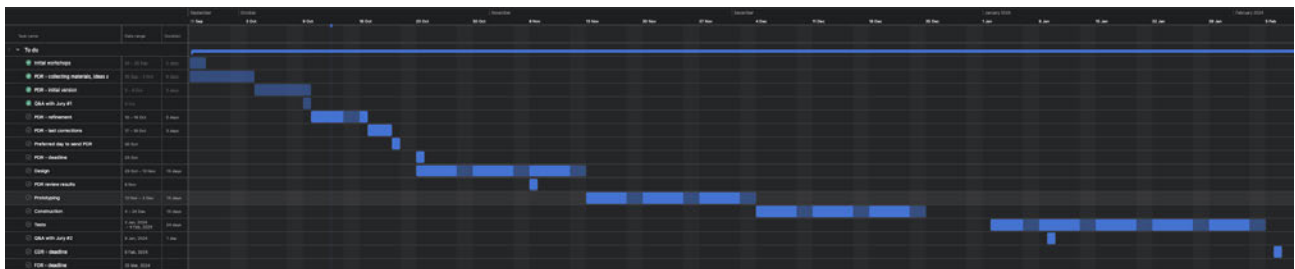
¹⁵ <https://cansat.esa.int/communicating-with-radio/>



3 PROJECT PLANNING

3.1 Time schedule

We decided to prepare PDR in 4 one week intervals, first was for collecting information and sources for the report, second was for writing the initial version of it, third for making it more detailed and the last one for last corrections. After PDR we plan to schedule work, on each phase of the project, in 3 week-long intervals. As of now we separated 5 main stages, namely: design, prototyping, construction, tests and writing CDR. They will consist of subparts. We make a rather strict schedule (tests should start before Q&A with Jury on 9 January 2024) to make sure that even if we fail to keep some dates we can take one more week to work better on some stage of the project. Materials to write CDR such as: notes, designs, etc. will be collected during all phases and consequently put together into the report. Gantt chart presenting our schedule until CDR (from Asana):



3.2 Task list

Tasks done before PDR:

- Writing PDR:
 - Sources and ideas collection
 - Primary and secondary mission factors description
 - Initial version of PDR
 - Refinement of PDR
 - Last corrections to PDR
- Outreach:
 - Instagram account setup and first posts
- Design of CanSat:
 - First 3D model of CanSat
 - Initial choice of components



Tasks to be done before CDR:

- Design:
 - 3D model of CanSat with all details
 - Parachute design
 - Ground station design
 - Electrical design
 - Detailed description of software
- Prototyping
 - Writing software
 - Checking if parts work alone
 - Making a parachute
- Construction
 - Assembling all parts into a CanSat
 - Mounting a parachute
 - Making ground station
- Tests
 - Primary mission sensors check
 - Secondary mission sensors and mechanism check
 - Parachute test
 - Troubleshooting
- Outreach
 - Weekly posts
- Budget:
 - Looking for sponsors

3.3 Resource estimation

3.3.1 Budget

Estimation of the costs:

- UV sensor - 6.50 EUR
- O2 sensor - 97.50 EUR
- Accelerometer, magnetometer - 19.90 EUR
- Elements from CanSat kit - 50 EUR
- motor - 24.50 EUR

These will be the most expensive parts. Even after adding other parts (filament, screws, etc.), the budget should not exceed 300 EUR.



4 OUTREACH PROGRAMME

Even though space exploration is a hot topic now, there are still a lot of people who do not know a lot about it. In our outreach programme we want to address them and people already fascinated about the subject, so we will create content appropriate for different groups of people. In order to achieve these goals we plan actions such as:

- regular posting on social media about progresses and project objectives (Instagram, Facebook),
- own website,
- meetings with other students group to talk about space exploration and our project,
- live transmission of CanSat descent during Launch Campaign,
- presenting project on science festivals

We plan to make posts about:

- CanSat competition
- Our primary and secondary mission
- Our team
- Progresses and achievements

So far we have designed our logo which is displayed on the first site of the document and we have set up Instagram account where we started posting about our project:

<https://instagram.com/exoscoutteam?igshid=MzRIODBiNWFIZA==>